

MAX ZHU

ATLANTA, GA | MAX.ZHU@GATECH.EDU | 404-940-3568 | [MAXZHU143.GITHUB.IO](https://maxzhu143.github.io)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S in Computer Engineering

Expected Graduation December 2026

- Threads: Systems and Architecture and Computing Hardware and Emerging Architecture

GPA: 3.9/4.0

TECHNICAL SKILLS

Software: Python, Java, C/C++, Javascript, HTML, Docker, CSS, AWS, MySQL, Django, React, Git, Ubuntu

Hardware: FPGA, Verilog, VHDL, Oscilloscopes, Digital Circuits, KiCad, Arduino, CAD

EXPERIENCE

CRNCH Rogues Gallery

Atlanta, GA

Reconfigurable Researcher

January 2025 – Present

- Using **Verilog** to create the **XChaCha20** encryption algorithm and **Docker** to deploy the package on a **PYNQ** board
- Offloaded cryptographic operations from the CPU to a **PYNQ FPGA**, seeing up to 10 times higher efficiency

MARGA Lab

Atlanta, GA

Wireless Digital Twin System Researcher

August 2024 – Present

- Created scripts in **Python** using **Sionna** to simulate and optimize transmitter-receiver placement by over 60%
- Analyzed and annotated 50+ research papers on **5G, Wi-Fi 6, ray tracing, wireless signals, and UAVs**
- Manipulated environments in **Blender** to reduce workload by 50% while maintaining over 95% accuracy.

Metro Healthcare

Vancouver, BC

Frontend Developer

May 2024 - August 2024

- Developed an interactive UI for 15+ pages with **React.js** to showcase clinical services and purchasable products
- Worked in an **Agile** project environment and continuously progressed by adapting to backend additions

PROJECTS

Spotify Car Thing and Wrapped App | Django, Python, HTML5, CSS, ESP8266, C++

- Created a DIY Spotify Car Thing replica with an **ESP8266**, **Spotify Developer API**, **C++**, and physical components
- Connected to the **Spotify Developer API** and used tokens for **OAuth** to grab live-playback and user song data
- Created custom profile analysis with **AI** and slide animations with **CSS** and **Javascript** for a personalized presentation
- Deployed the website with **Heroku** handling pipeline conflicts while securely handling user data

Atlanta Food Finder | Django, Python, HTML5, CSS, MySQL, Git, Herokuapp

- Worked as a **scrum master** and **full-stack developer** to handle branch, merge, and version conflicts while delegating tasks and overcoming obstacles to create a **Django web application**
- Implemented and scraped the **Google Maps API** to display nearby restaurant location and information
- Utilized **MySQL** to manage user data including usernames, passwords, favorites, and reviews

Portfolio Website | maxzhu143.github.io

- Designed and developed an engaging website using **HTML**, **CSS**, and **JavaScript** to showcase projects

Cloud Server | Linux CLI, Docker

- Configured and deployed a cloud server with **Ubuntu Server** to replace multiple personal subscription services
- Exclusively utilized **Linux CLI** (command-line interface) for system administration, server organization, and **automation**

EXTRACURRICULARS

Silicon Jackets | Digital Designer and Verification Engineer

August 2024 – Present

- Designed and implemented an **arithmetic logic unit** for a CPU from scratch with **Verilog**
- Developed and synthesized over 100+ tests with **Python** to verify correct hardware functionality

HIVE | Master Peer Instructor

December 2023 – Present

- Led workshops and guided users laser cutting, 3d printing, wood working, and **PCB fabricating**
- Fixed and maintained sewing, PCB creating, and wood working machines,